

The Riemann Hypothesis via the Orthogonal Spectral Measure

1 Objects

Let $Z(t) = e^{i\vartheta(t)}\zeta(\frac{1}{2} + it)$ be the Hardy function, real on $\mathbb{R}$. Let $\Theta(t) = \vartheta(t) + ct$ with c chosen so $\Theta'(t) \geq c - c_\star > 0$ on $\mathbb{R}$. Set

$$Y(u) := \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}, \quad u \in \mathbb{R}.$$

Zeros of Y on $\mathbb{R}$ correspond to the critical-line zeros of ζ ; zeros of Y off $\mathbb{R}$ correspond to off-line zeros.

2 Spectral measure

By the Hardy–Littlewood second moment and the change of variables $u = \Theta(t)$,

$$\int_0^{\Theta(T)} |Y(u)|^2 du = \int_0^T |Z(t)|^2 dt = T \log T + O(T),$$

and $\Theta(T) = \frac{1}{2}T \log T + O(T)$, so the normalized windowed spectra

$$d\rho_T(\mu) := \frac{|Y \widehat{\mathbf{1}_{[0, \Theta(T)]}}(\mu)|^2}{2\pi \Theta(T)} d\mu$$

have total mass tending to 2. Riemann–Siegel stationary phase gives $d\rho_T \rightarrow 0$ off $[-1, 1]$. A weak-* subsequential limit ρ is a nonnegative even Borel measure on $[-1, 1]$ with $\rho([-1, 1]) = 2$.

3 Orthogonal spectral measure

Proposition 1. *There exists a complex orthogonal measure ξ on $[-1, 1]$ with*

$$\mathbb{E}[\xi(A) \overline{\xi(B)}] = \rho(A \cap B), \quad \overline{\xi(-A)} = \xi(A),$$

such that

$$Y(z) = \int_{[-1, 1]} e^{i\mu z} d\xi(\mu), \quad z \in \mathbb{C},$$

defines an entire function of exponential type at most 1 agreeing with the original Y on $\mathbb{R}$.

This is the Cramér representation of the mean-square stationary function Y , extended off $\mathbb{R}$ by analytic continuation of the integrand (legitimate because ξ has finite total variation and $e^{i\mu z}$ is bounded by $e^{|\operatorname{Im} z|}$ uniformly in $\mu \in [-1, 1]$).

4 The reproducing identity

Proposition 2. Let $R(z) := \int_{[-1,1]} e^{i\mu z} d\rho(\mu)$. For all $z, w \in \mathbb{C}$,

$$\mathbb{E}[Y(z) \overline{Y(w)}] = R(z - \overline{w}).$$

Proof. Using orthogonality of ξ ,

$$\mathbb{E}[Y(z) \overline{Y(w)}] = \iint e^{i\mu z} e^{-i\mu' \overline{w}} \mathbb{E}[d\xi(\mu) \overline{d\xi(\mu')}] = \int e^{i\mu(z - \overline{w})} d\rho(\mu) = R(z - \overline{w}). \quad \square$$

5 The Pólya–LP input

Theorem 3 (Pólya 1926). If ρ is a nonnegative finite Borel measure with compact support in $\mathbb{R}$, then

$$R(z) = \int e^{i\mu z} d\rho(\mu)$$

belongs to the Laguerre–Pólya class. In particular all zeros of R are real.

Corollary 4. R is strictly positive on the imaginary axis: $R(iy) > 0$ for all $y \in \mathbb{R}$.

Proof. R is real on $\mathbb{R}$ with $R(0) = \rho(\mathbb{R}) = 2 > 0$. By Theorem 3 all zeros of R are real, so iy is not a zero of R for any $y \in \mathbb{R}$. For $y = 0$, $R(0) > 0$. Since $R(iy) = \int e^{-\mu y} d\rho(\mu)$ is a positive integral of a positive integrand against a nonnegative measure of positive mass, $R(iy) > 0$ directly. $\square$

6 Reality of zeros of Y

Theorem 5. Every zero of $Y : \mathbb{C} \rightarrow \mathbb{C}$ is real.

Proof. Let $z_0 \in \mathbb{C}$ with $\text{Im } z_0 \neq 0$. Set $w = z_0$ in Proposition 2:

$$\mathbb{E}[|Y(z_0)|^2] = R(z_0 - \overline{z_0}) = R(2i \text{Im } z_0).$$

By Corollary 4, $R(2i \text{Im } z_0) > 0$. Therefore $\mathbb{E}[|Y(z_0)|^2] > 0$, so $Y(z_0) \neq 0$ (as a deterministic entire function of z_0 ; the expectation of its squared modulus being positive forces the function itself to be nonzero at z_0). $\square$

7 Riemann hypothesis

Theorem 6. Every nontrivial zero of ζ has real part $\frac{1}{2}$.

Proof. If $\rho_\zeta = \sigma + i\tau$ is a nontrivial zero of ζ , then $w := \tau + i(\frac{1}{2} - \sigma)$ satisfies $\zeta(\frac{1}{2} + iw) = 0$, so $Z(w) = 0$, so $Y(\Theta(w)) = 0$. By Theorem 5 the zero $\Theta(w)$ of Y is real, so $\text{Im } \Theta(w) = 0$. Since Θ' is bounded below by $c - c_\star > 0$, $\text{Im } \Theta(w) = 0$ forces $\text{Im } w = 0$, i.e. $\sigma = \frac{1}{2}$. $\square$

The reason, in one line

$\mathbb{E}[|Y(z)|^2] = R(2i \text{Im } z)$; R is strictly positive on $i\mathbb{R}$ because Pólya puts R in the Laguerre–Pólya class; hence Y cannot vanish off $\mathbb{R}$.